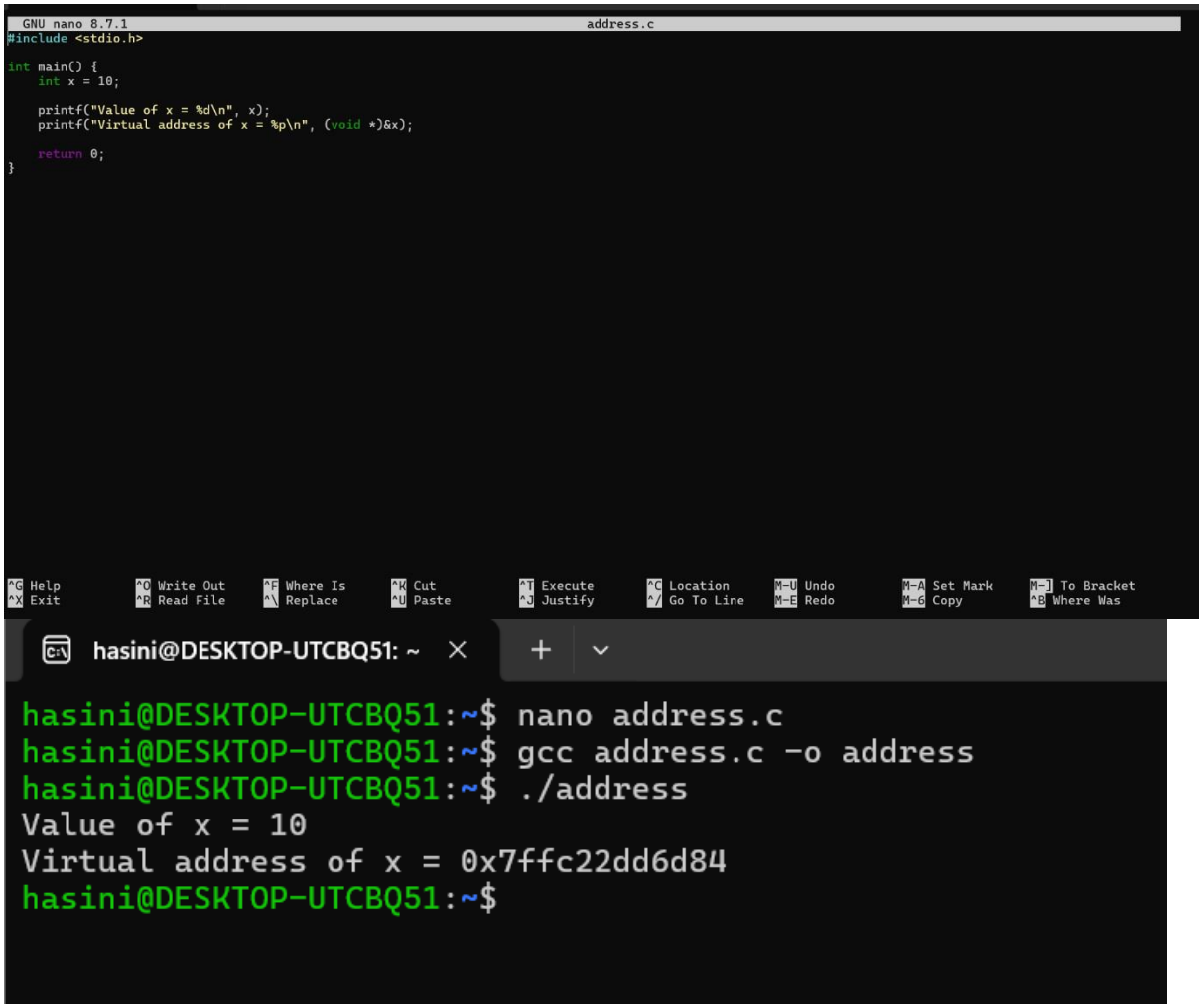
Name: J.Chandra Hasini

RollNo:2520030086

OSSP PRACTICAL

1.Virtual Address:



2. Demand paging:

hasini@DESKTOP-UTCBQ51: ~

GNU nano 8.7.1 page.c *

```c
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

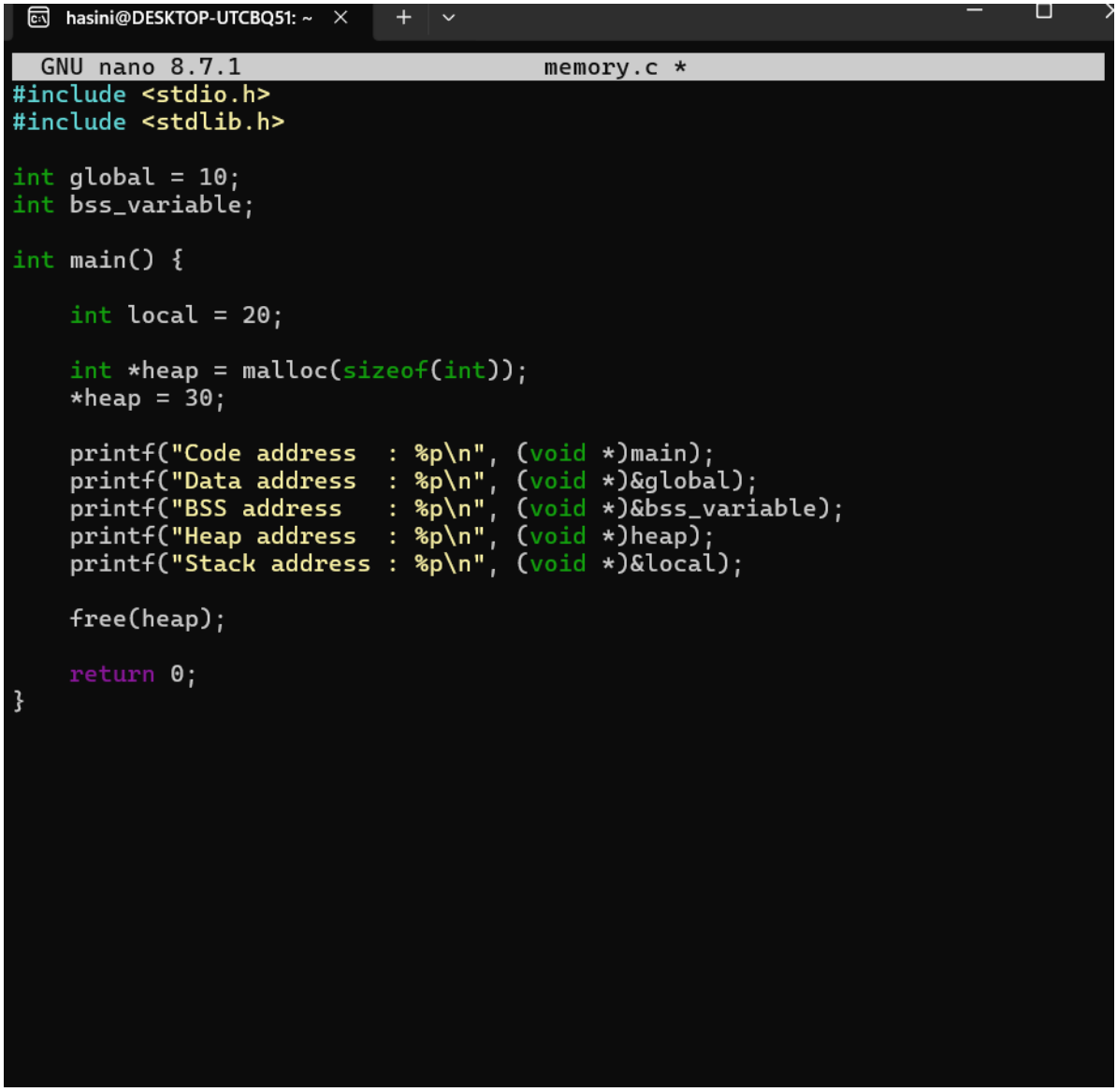
    return 0;
}
```

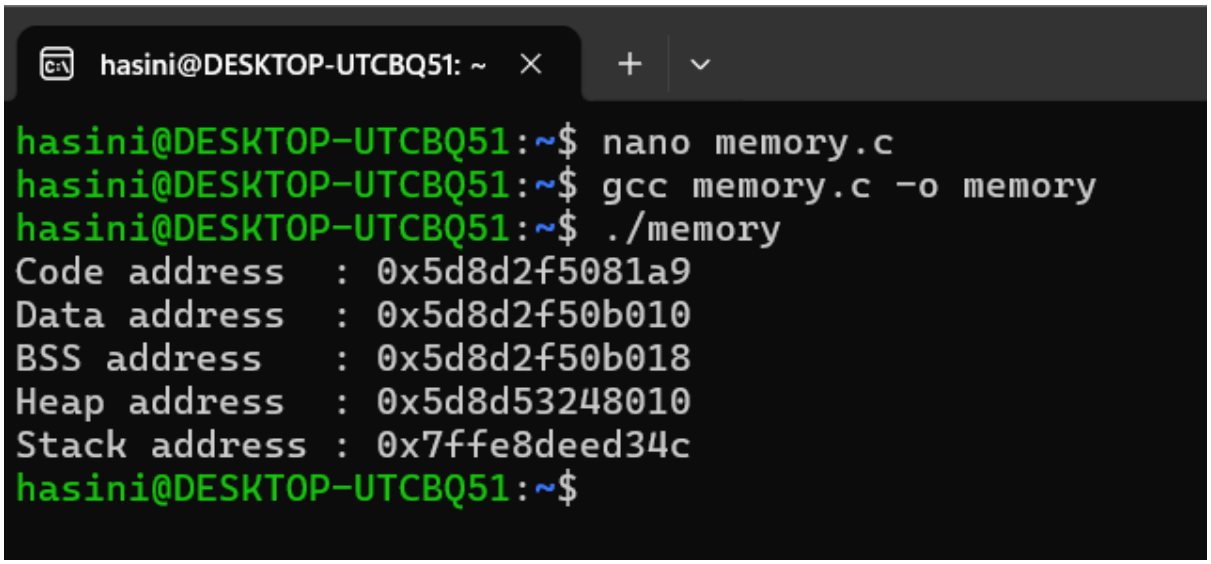
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

hasini@DESKTOP-UTCBQ51: ~

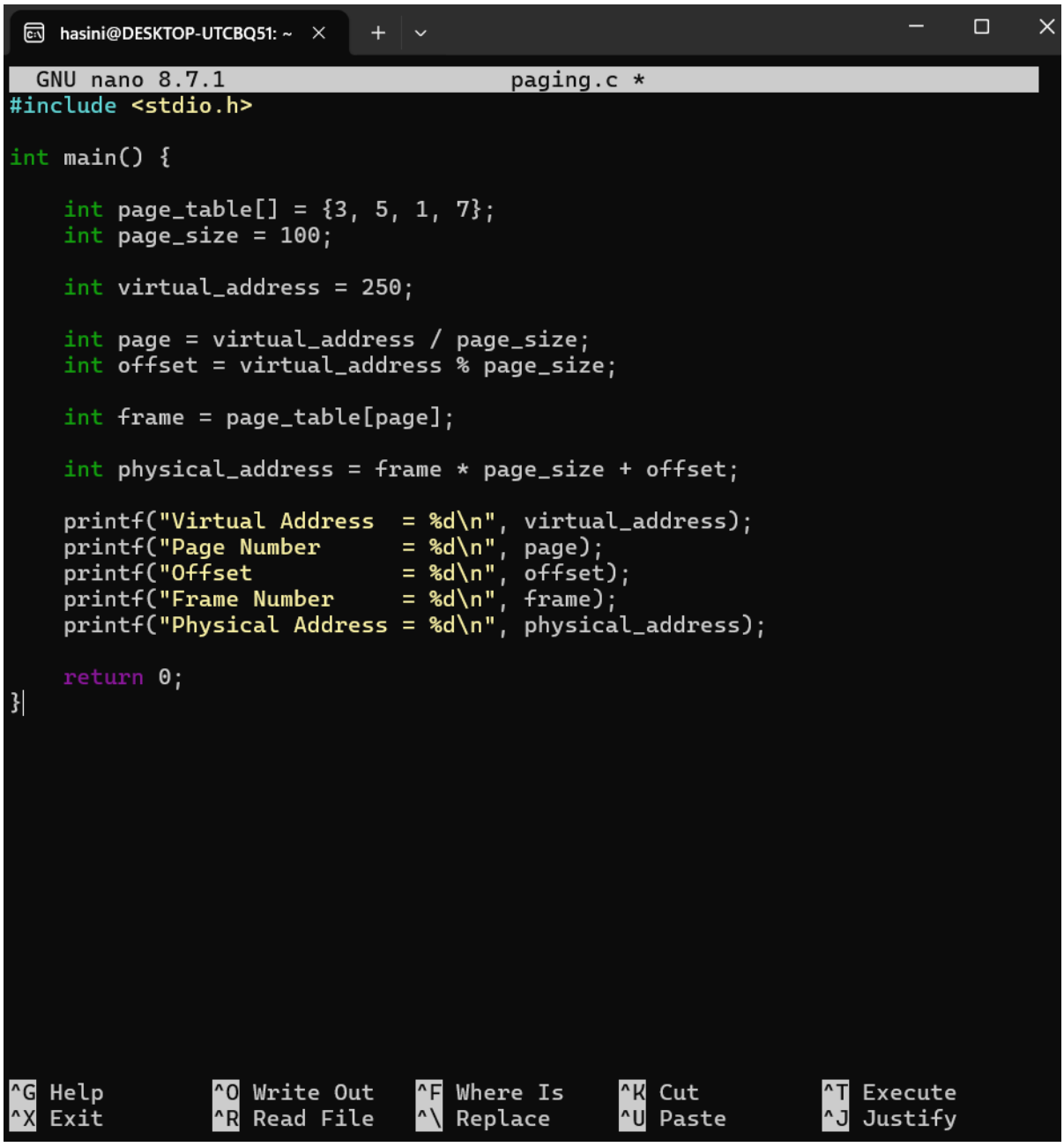
```
hasini@DESKTOP-UTCBQ51:~$ nano page.c
hasini@DESKTOP-UTCBQ51:~$ gcc page.c -o page
hasini@DESKTOP-UTCBQ51:~$ ./page
Enter page number (0-3): 5
Page is already in memory.
Physical frame = 0
hasini@DESKTOP-UTCBQ51:~$
```

3. process spaces:




hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ nano memory.c
hasini@DESKTOP-UTCBQ51:~$ gcc memory.c -o memory
hasini@DESKTOP-UTCBQ51:~$ ./memory
Code address : 0x5d8d2f5081a9
Data address : 0x5d8d2f50b010
BSS address : 0x5d8d2f50b018
Heap address : 0x5d8d53248010
Stack address : 0x7ffe8deed34c
hasini@DESKTOP-UTCBQ51:~$

4. adress translation:



```
hasini@DESKTOP-UTCBQ51:~$ nano paging.c
hasini@DESKTOP-UTCBQ51:~$ gcc paging.c -o paging
hasini@DESKTOP-UTCBQ51:~$ ./paging
Virtual Address  = 250
Page Number      = 2
Offset           = 50
Frame Number     = 1
Physical Address = 150
hasini@DESKTOP-UTCBQ51:~$
```

5. Malloc and free

```
 GNU nano 8.7.1                    malloc.c *
#include <stdio.h>
#include <stdlib.h>

int main() {

    int *p;

    p = malloc(5 * sizeof(int));

    if (p == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

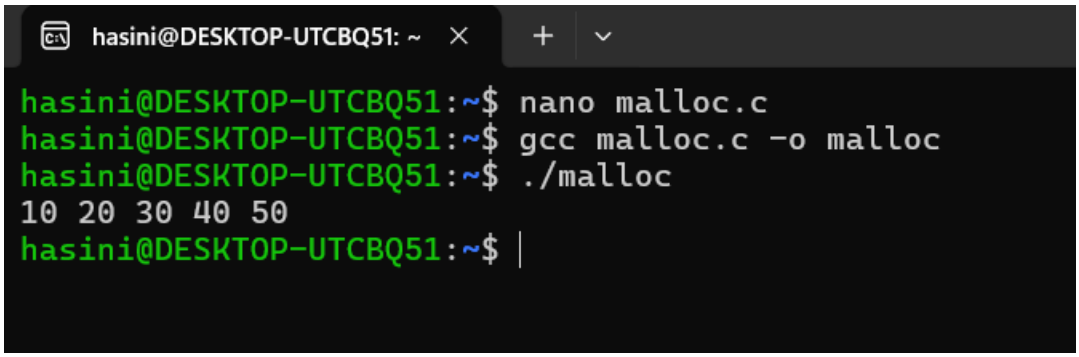
    for (int i = 0; i < 5; i++) {
        p[i] = (i + 1) * 10;
    }

    for (int i = 0; i < 5; i++) {
        printf("%d ", p[i]);
    }

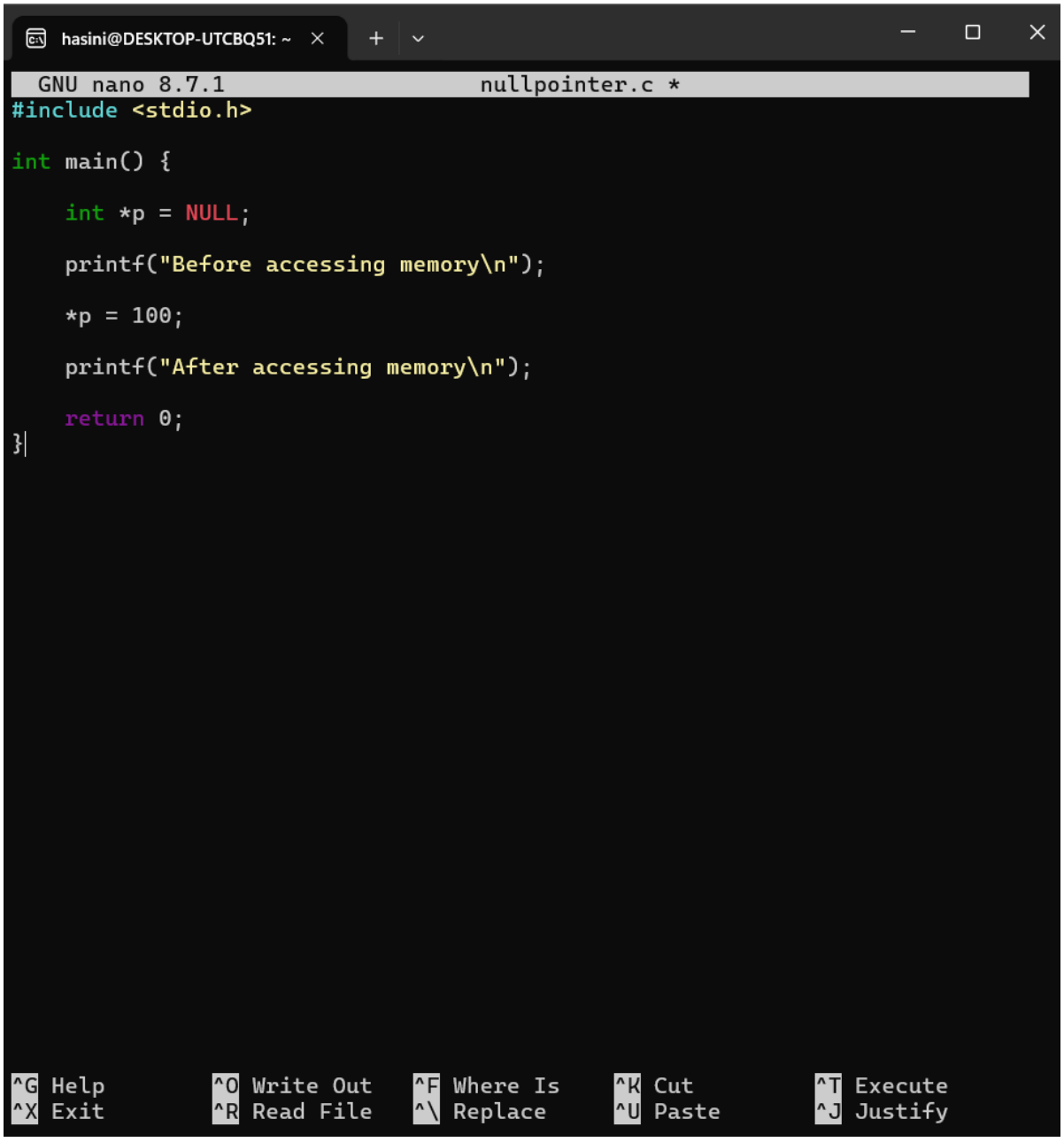
    printf("\n");

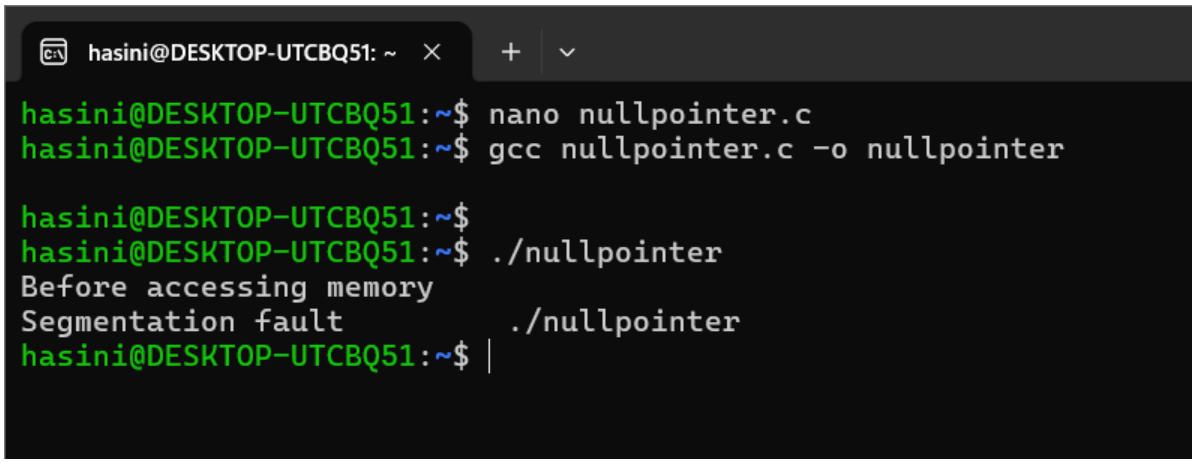
    free(p);

    return 0;
}
```



6. Segmentation Fault:




hasini@DESKTOP-UTCBQ51: ~
hasini@DESKTOP-UTCBQ51:~$ nano nullpointer.c
hasini@DESKTOP-UTCBQ51:~$ gcc nullpointer.c -o nullpointer
hasini@DESKTOP-UTCBQ51:~$
hasini@DESKTOP-UTCBQ51:~$ ./nullpointer
Before accessing memory
Segmentation fault ./nullpointer
hasini@DESKTOP-UTCBQ51:~$